



Inclusive Mobility: Dismantling transportation injustice at the intersection of race and disability

Erica Twardzik^{a,b,c,d,*}, Michael R. Desjardins^{a,e}, Frank C. Curriero^{a,e},
Bonnielin K. Swenor^{a,d,f,g}, John W. Jackson^{a,h}, Jennifer A. Schrack^{b,c},
Keshia M. Pollack Porter^{h,i,j}

^a School of Kinesiology, University of Michigan, Ann Arbor, MI, 48109, USA

^b Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, 21205, USA

^c Center on Aging and Health, Johns Hopkins University, Baltimore, MD, 21205, USA

^d Disability and Health Research Center, Johns Hopkins University, Baltimore, MD, 21205, USA

^e Spatial Science for Public Health Center, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, 21205, USA

^f School of Nursing, Johns Hopkins University, Baltimore, MD, 21205, USA

^g Wilmer Eye Institute, Johns Hopkins School of Medicine, Johns Hopkins University, Baltimore, MD, 21205, USA

^h Johns Hopkins Center for Health Disparities Solutions, Baltimore, MD, 21205, USA

ⁱ Department of Health Policy and Management, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, 21205, USA

^j Johns Hopkins Institute for Health and Social Policy, Baltimore, MD, 21205, USA

ARTICLE INFO

Keywords

Transportation injustice
Health equity
Intersectionality

ABSTRACT

Transportation impacts population health. Historical trauma, structural inequities, and institutional discrimination have created transportation injustice. Transportation injustice is a product of systemic racism and ableism which perpetuates inequities, discrimination, and exclusion. However, systemic racism and ableism can compound injustice given one's social identities. In aligning with the principles of mobility justice and Crip Mobility Justice, this paper asserts that an intersectional lens is needed to dismantle transportation injustice and create a sustainable transportation system rooted in health equity. Specifically, social identities do not exist independent of each other, creating a complex convergence of oppression in transportation access. To support this viewpoint, we: (1) describe transportation history among two overlapping historically marginalized populations in the United States, Black people and people with disabilities, (2) articulate the impact transportation injustice has had on public health, and (3) advocate for an intersectional lens to dismantle unjust systems, policies, and structures.

Transportation injustice is defined as the structural conditions that perpetuate inequities, discrimination, and exclusion in transportation systems, disproportionately affecting marginalized communities based on factors such as disability, race, class, gender, and age.¹ Transportation is a modifiable target to achieve health equity and directly impacts population health through active transport, access to goods and services, and social participation. Health equity “means reducing and ultimately eliminating disparities in health and in the determinants of health that adversely affect excluded or marginalized groups”.² Public transit systems frequently design their services with a specific “ideal rider” in mind, usually an economically secure, able-bodied, white male commuter.³ This bias can exclude and disadvantage individuals who don't match this profile, by failing to meet their needs. Transportation

justice-oriented research aims to identify and dismantle the entrenched structural factors that enable and sustain inequities and discrimination.⁴ Although the United States (U.S.) Department of Transportation launched the Justice40 initiative in 2021 to address decades of underinvestment in disadvantaged communities, it has since been rescinded by a new federal administration.⁵ Federal programs, such as the All Station Accessibility Program (ASAP), which previously operated under guidance to direct 40 % of benefits to disadvantaged communities, now function without these explicit equity requirements. While ASAP continues to exist as it was established through the Bipartisan Infrastructure Law, the removal of Justice40 requirements may impact how benefits are distributed across communities. To create a sustainable transportation system and achieve health equity despite these policy changes,

* Corresponding author. SKB 3387 830 North University Ann Arbor, MI48109-1048, USA.

E-mail address: etwardzi@umich.edu (E. Twardzik).

<https://doi.org/10.1016/j.dhjo.2025.101848>

Received 1 October 2024; Received in revised form 9 April 2025; Accepted 5 May 2025

Available online 6 May 2025

1936-6574/© 2025 Published by Elsevier Inc.

transportation planning efforts must be centered on people who are most impacted and apply an intersectionality framework in program implementation.⁶

Intersectionality recognizes that individuals often experience multiple, overlapping forms of discrimination and/or privilege based on their various social identities.⁷ Dismantling transportation injustice and mitigating its impact on health hinges on the use of intersectionality theory. Applying an intersectionality framework facilitates a more nuanced, accurate, and actionable understanding of transportation injustice. Furthermore, representatives from the National Institutes of Health assert it is “imperative that health researchers embrace an intersectional lens” when “quantifying social inequalities that lead to health disparities”.⁸ Intersectionality is necessary for the development of sustainable transportation systems that enhance health equity in the population.

In this article, we identify existing structural conditions that perpetuate transportation injustice in the U.S. among two overlapping historically marginalized populations – non-Hispanic Black people (hereinafter referred to as Black) and people with disabilities. Using these two social identities as an example we examine the intersectionality of race and disability in the context of transportation injustice. We argue that the marginalization of Black and disabled people compounds their disadvantage, creating unique and amplified experiences of transportation inequity. By identifying existing structural conditions that perpetuate transportation injustice among these two intersecting populations, we aim to articulate the impact of transportation injustice on public health and identify strategies for dismantling transportation injustice.

1. Transportation injustice context

Injustice – unfair treatment : a situation in which the rights of a person or a group of people are ignored⁹

1.1. Race-based injustice

Racist policies and systems have built and maintained inequities in the U.S. transportation system.⁹ For Black Americans, transportation injustice began before stepping foot on American soil. Forced transportation of slaves to the American colonies and oppression after arrival (e.g., 1850 Fugitive Slave Act) subsumed travel rights based on skin color. The continuation of legalized slavery, which is at the crux of structural racism, and transportation disenfranchisement occurred for more than two centuries. After the abolition of slavery, Black Americans continued to be stripped of transportation rights through segregation, terrorism, and discrimination. The Supreme Court upheld segregated transportation systems until *Morgan v. Virginia* (1946) and *Boynton v. Virginia* (1960) which ruled that segregated public buses were unconstitutional. Following the Supreme Court’s ruling, transportation systems continued to be platforms for state permitted terrorism and discrimination. Exemplified by coordinated efforts between the police department and the Ku Klux Klan to attack Freedom Riders, civil rights activists who rode interstate buses in mixed racial groups across the Deep South.¹⁰ Freedom riders encountered horrific situations including arrests, physical attacks, and threats to their lives.

In parallel with overt transit disenfranchisement, U.S. laws, policies, and infrastructure manufactured racial residential segregation. Racial residential segregation is a known cause of health disparities.¹¹ In the 1930s the Federal Housing Administration, operated through the New Deal, restricted Black families’ access to capital while increased that of non-Hispanic white (hereinafter referred to as white) families.¹² Furthermore, federal policies established and reinforced segregation in the U.S. by refusing to insure mortgages in predominantly Black neighborhoods, a policy known as “redlining”.¹² This encouraged white families to abandon inner cities in exchange for suburban living and

prevented ethnic minorities from doing the same. “White flight” caused and contributed to urban decay and concentration of poverty and disrepair within the urban core. Following urban decay came urban renewal projects and public housing construction, which further disenfranchised Black Americans.¹³ Interstate highway system investments displaced Black households and cut through thriving Black neighborhoods, destroying homes, churches, schools, and businesses in its expansion.¹⁴ Gentrification and a shift toward automobile transportation led to the displacement of Black families to suburban dwellings desolate of transit access,¹⁵ defined as affordable, available, adjacent, accommodating, and acceptable.¹⁶

Racism continues to shape long lasting transportation decisions and impact the health of the population. For example, Baltimore City has been described as a “Black Butterfly” - contrasting the affluent, majority white neighborhoods in Inner Harbor stretching north in Baltimore City (the “white L”) with the low-income, majority Black neighborhoods of East and West Baltimore (the “Black Butterfly”).¹⁷ Previous research has found the greatest public transportation stops along the “white L”.¹⁸ Conversely the expansive areas of the “Black Butterfly” host the fewest public transportation stops, greatest need of transportation investment, and highest disease prevalence.¹⁸ Maryland proposed the Red Line to be an east-west transportation line providing essential access for predominantly Black populations. The Red Line project was granted federal funding and roughly \$300 million was completed in planning, design, and land acquisition for the project.¹⁹ However, in 2015 Maryland administration pulled funds from the project and allocated funding for a Purple Line which benefited a predominantly white population in the Maryland suburbs.²⁰ The administration did not seek input from the Department of Transportation, Baltimore citizens, or community groups on this decision, which impacted a largely public transportation dependent population.²⁰

1.2. Disability-based injustice

Ableism is defined as “a system of assigning value to people’s bodies and minds based on a societally constructed ideas of normalcy,” which shapes an individual’s worth.²¹ Ableism is pervasive within U.S. culture. When cities developed and invested in public transportation systems, people with disabilities were not included as planners or thought of as potential users. Disabled populations were marginalized and denied access to public transportation well into the 20th century. Inaccessible, and thereby exclusionary, infrastructure persists across the U.S. today. The physical environment, including vehicles, facilities, and pedestrian streetscapes, creates transportation disenfranchisement for the disability community, which impacts quality of life, physical activity, and participation.²² Despite the passage of section 504 of the Rehabilitation Act in 1973 (prohibiting disability discrimination in federally funded programs), public transportation systems continue to be inaccessible. In 1978 the Gang of 19 protested the inaccessible transportation system by positioning their bodies in front of buses while chanting “We Will Ride!”²³ These protests highlighted the absence of wheelchair-accessible buses across the country and was the origin of the grassroots disability rights organization *Americans Disabled for Accessible Public Transit* (ADAPT). ADAPT expanded quickly and is a major U.S. disability advocacy group.²⁴ Today, fifty years after the passage of section 504, nearly two-thirds of people with disabilities report problems using public transportation.²⁵ Furthermore, 70 % of disabled adults reduce daily travel because of inaccessible transportation.²⁶ These barriers create systemic exclusion of disabled people.

To address systemic exclusion of people with disabilities, the Americans with Disabilities Act (ADA) prohibits discrimination based on disability. ADAPT was a key advocacy group ensuring that transportation provisions were included in the ADA.²⁷ However, transportation systems adoption of ADA requirements has been disparate due to voluntary compliance and freedom to deny accommodations if financially burdensome.²⁸ While the ADA requires public entities

provide complementary demand-response services when fixed route services are inaccessible, there are significant barriers to this approach. The segregated demand-response system is challenging for transportation systems to operate due to financial and structural reasons.²⁹ This thereby impacts the availability, quality, quantity, and flexibility of public transportation services available to the disability community. Paratransit trips are often less efficient, reliable, and flexible than comparable fixed-route transportation options.³⁰ Furthermore, labor shortages have resulted in major scale backs to paratransit coverage areas and denial of trip requests.^{31,32} An investigation by Access Living found 41 % of pickup times in Chicago’s paratransit system were late, among which nearly half were over 40 minutes late.³³ A shift in discussion from ADA compliant transit towards addressing the social and political structures oppressing disabled people is desperately needed.³⁴ Crip Mobility Justice is a framework which prioritizes accessibility for disabled people, and builds for the most marginalized among them.³⁵ This framework states that greater commitments to interdependence and widespread accessibility are needed.³⁵ Crip Mobility Justice includes greater flexibility in multi-model transit systems and also prioritizes “curb cuts, adequate and sensory-friendly lighting, spaces of respite and quiet, public restrooms and water fountains, as well as housing justice and the abolition of policing and surveillance.”³⁵

1.3. Intersectionality of race- and disability-based injustice

For Black people with disabilities, experiences of transportation injustice can be amplified. The sections above describe how Black people and people with disabilities face barriers and unequal access to public transportation and their implications for health. However, less is known about transportation experiences among people who identify as belonging to both groups. Even though a greater proportion of Black people have a disability compared with white people in the U.S. (Table 1).³⁶ Table 1 displays the proportion of people in the U.S. with a disability by age group and race. At every age, Blacks have a higher proportion of disability than whites. Additionally, adults aged 65 and older (the group with the highest proportion of disability) represent a much smaller proportion of Blacks (12.5 %) compared with whites (20.2 %), due to early mortality. Meaning that the total proportion of Blacks with disability is much lower than it otherwise would be, if not for existing health disparities.

The dearth of evidence examining the intersection between ableism and racism, stem from disability studies typically excluding racial considerations, while race studies tend to omit disability perspectives.³⁷ Historically, justice movements have also not been inclusive of people with other marginalized identifies. For example, disability activism in the 1970s often excluded members of communities of color based in fear that racial identity would divide people with disabilities from one another.³⁸ However, the compounding effects of racism and ablism is experienced and shared by Black disabled riders. For example, the #WheresMyRide campaign highlighted failures of the paratransit

Table 1
Proportion of people with a disability in the United States in 2023, stratified by Black and white race using the American Community Survey’s 5-year estimates.

	Black		White	
	Proportion of Black population	Proportion of people with disabilities	Proportion of white population	Proportion of people with disabilities
Under 18 years	24.9 %	5.6 %	19.6 %	4.5 %
18–64 years	62.6 %	13.4 %	60.2 %	10.9 %
65+ years	12.5 %	37.3 %	20.2 %	32.1 %
Total	–	13.9 %	–	14.5 %

service in Baltimore, Maryland. As part of this campaign, Tierra Flecther shared her experience and a Black, disabled woman being stranded by the paratransit system.

“I scheduled it [My ride] seven days in advance per the request of Mobility. I heard my phone ringing. They said we’re not going to be able to pick you up until 2:00 AM in the morning. This was at 8:55. I’m in an area where it’s darkly lit. Anything can happen to me. So, then that caused me to have a panic attack.”

Unreliable transportation and no access to backup vehicles resulted in a panic attack, likely due to feelings vulnerability and concerns for her safety. Her lived experience reflects U.S. environments, where disabled people of color are at risk of police targeting, and up to 50 % of all people killed by law enforcement are disabled.³⁹ Her experience would have been very different if she was in a different social, natural, physical, and/or political environment.

Individually and combined, racism and ableism may reinforce transportation injustice in the planning process, community engagement process, and services of provisions. For example, race contributes to skepticism of having a legitimate disability.⁴⁰ This could impact Black people with disabilities eligibility for demand-response transportation services. Eligibility determination often includes an assessment, and racial profiling could negatively impact provision of services. This is especially concerning given the higher demand for paratransit services from neighborhoods with a higher proportion of racial minorities.⁴¹ At the organizational level, disinvestment in public transit infrastructure within communities of color may result in inadequate vehicles, audio/visual communication, or travel training programs.

2. Transportation injustice – A manufactured public health crisis

Transportation environments and public health are inextricably linked, with suboptimal urban and transport planning causing morbidity and premature mortality.⁴² Ableism and racism have created transportation injustice and manufactured a public health crisis. There are several mechanisms through which transportation planning impacts public health, including exposure to environmental hazards (e.g., air pollution, air quality),⁴³ traffic related injuries,^{44,45} and lifestyle physical activity.⁴⁶ Furthermore, among public transportation users, being non-white and having a physical disability was associated with deferring medical care during the COVID-19 pandemic.⁴⁷ Not only is transportation itself a social determinant of health, but it also links to almost every other social determinant of health. Social determinants of health are major contributors to health outcomes, including length of life and quality of life.⁴⁸ Further, public transportation is a much safer way to travel. Traveling by public transit results in less than 10 % per-mile traffic casualty (injury or death) compared with traveling by automobile.⁴⁹

Who and what populations benefit from transportation improvements is an issue of transportation justice. Mobility justice calls for transformative policies that ensure equitable access to transportation as a fundamental human right, thereby improving health outcomes.⁵⁰ Public health has a crucial role in investigating, monitoring, and mitigating health inequities. This work must begin with a critical reflection on how public health has historically perpetuated racism and ableism.

3. Strategies to dismantle transportation injustice

Dismantling transportation injustice requires that everyone has access to transportation that allows for a meaningful, dignified, and fulfilling life.⁵¹ Steps to dismantling transportation injustice include naming it exists, asking how it operates, and strategizing to act.⁵² By acknowledging and first naming ableism and racism as one of the root causes of the inequities in transportation systems, we can then investigate which mechanism(s) it is operating, and dismantle the systems,

policies, and structures that created the inequities. Within public health, we will need to examine existing social structures, systems, and norms that contribute to transportation injustice. This will provide greater awareness, reflection, and dialogue, which can then stimulate action and transformation.⁵³ As a research community, greater attention to compounding effects of systemic racism and ableism on transportation and health is needed. Within this viewpoint we present a conceptual framework to guide future research questions connecting racism and ableism to public transportation provisions and public health implications (Fig. 1). Research has an important role to play in setting the narrative of transportation injustice as a fundamental cause of health inequities.

Dismantling transportation injustice will require policy and system changes that intentionally and explicitly challenge and address existing racist and ableist systems. To dismantle transportation injustice, we must have information about disability in all locations where demographic information is collected. Without information about disability we cannot assess, plan, and evaluate the impact of public transportation injustice. Data about neighborhood composition can inform approaches like catalytic forecasting to dismantle transportation

injustice.¹⁵ Catalytic forecasting can be used to more accurately justify need for increased transportation access, lines, frequency, and accessibility, and can assist in the location of transportation stops, thereby increasing overall efficiency and equity of movement in cities.¹⁵ However, this is not possible without robust information about race and disability within transportation and health systems.

Racism and ableism operate through similar but unique mechanisms which interact and reinforce one another; however, effects on people who identify as both Black and disabled are not well understood. Intersectionality describes the way that systems of power and oppression interact to form an individual's unique experience.⁷ People "do not live single issue lives," and what is known about transportation experiences for Black, disabled people is limited.⁵⁴ Examining the role of transportation systems on intersecting identities is critical to understanding transportation injustice. Characterizing, measuring, and dismantling transportation injustice must include leadership by those who are most impacted.⁶ Public systems must empower voices that have been sidelined in transportation planning and decision-making efforts.^{34,55} Embracing a mobility justice approach necessitates community-led planning and policy-making that center the voices of marginalized

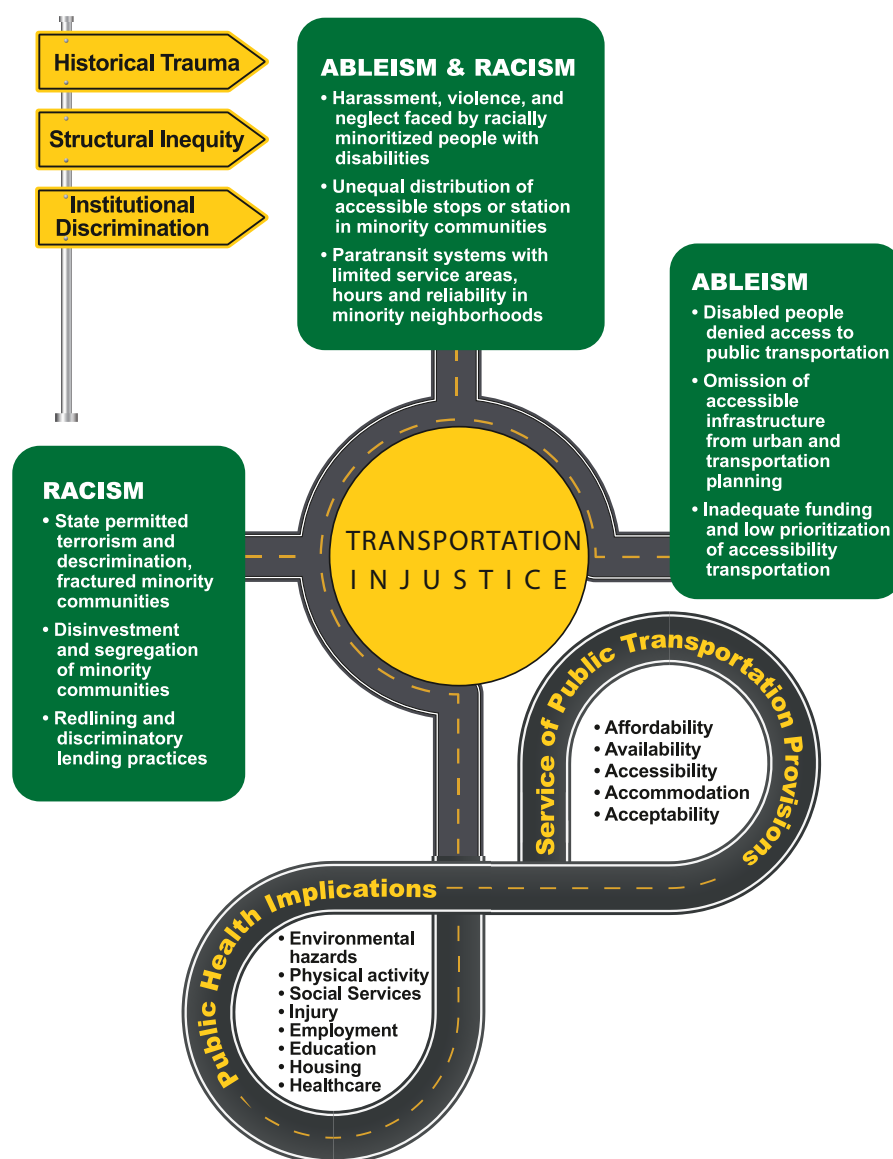


Fig. 1. Transportation injustice: Intersections of racism and ableism in United States public transportation systems shaping services of provisions and population health.

groups.⁵⁰ Meaningful public involvement in transportation decision-making is a key tenant of transportation planning and is required by all Federal Transit Administration (FTA) grant recipients.⁵⁶ Drawing upon the expertise from Black and disabled communities as leaders and colleagues in the transportation process can mitigate unintended consequences. To accomplish these goals public engagement must be accessible, offer flexible opportunities for feedback, and share power so community voices are part of decision-making. Removing institutional barriers includes developing policies that legitimize public involvement, dedicating budget and staff to these efforts, and partnering with communities to improve decision making.⁵⁷

The process of engaging historically marginalized communities can also be amplified to address structural racism and ableism. The FTA guidance on implementing Title VI of the Civil Rights Act of 1964 requires that transportation agencies enact a “major” change to their service, they conduct an analysis to show that the change does not have a disparate impact on people of color.⁵⁸ To end transportation injustice, this regulatory policy could instead boldly require that all changes (not only major changes) be viewed through an anti-racist and anti-ableist lens. Involving potential users at all stages of the transportation life-cycle (design, advisory boards, procurement, and implementation) could mitigate unintended consequences.

To achieve equity in public transportation and public health, we need to acknowledge the roles of historical trauma, structural inequity, and institutional discrimination driven by racism and ableism which created the current transportation landscape in the U.S. Reframing the narrative toward transportation injustice is a first step towards dismantling existing systems, policies, and structures to promote transit and health equity.

CRediT authorship contribution statement

Erica Twardzik: Writing – original draft, Visualization, Validation, Funding acquisition, Conceptualization. **Michael R. Desjardins:** Writing – review & editing, Investigation. **Frank C. Curriero:** Writing – review & editing, Investigation. **Bonnielin K. Swenor:** Writing – review & editing, Investigation. **John W. Jackson:** Writing – review & editing, Investigation. **Jennifer A. Schrack:** Writing – review & editing, Investigation. **Keshia M. Pollack Porter:** Writing – review & editing, Investigation, Conceptualization.

Funding statement

Research reported in this publication was supported by the National Institute on Aging of the National Institutes of Health under Award Number R00AG081563. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Conflicts of interest

None to disclose.

References

- Pereira RH, Karner A. *Transportation Equity*. Elsevier; 2021.
- Braveman P, Arkin E, Orleans T, Proctor D, Acker J, Plough A. What is health equity? *Behavioral Sci. Policy*. 2018;4(1):1–14.
- Lubitow A, Rainer J, Bassett S. Exclusion and vulnerability on public transit: experiences of transit dependent riders in Portland, Oregon. *Mobilities*. 2017;12(6): 924–937.
- Karner A, Bills T, Golub A. *Emerging Perspectives on Transportation Justice*. Elsevier; 2023, 103618.
- U.S. Department of Transportation. *Justice40 Initiative*; 2024. . Accessed May 7, 2024.
- Sins Invalid. 10 principles of disability justice. <https://www.sinsinvalid.org/blog/10-principles-of-disability-justice>; April 21, 2023, 2015.
- Crenshaw K. *Demarginalizing the intersection of race and sex: a black feminist critique of antidiscrimination doctrine, feminist theory and antiracist politics*. *Feminist Legal Theories*. Routledge; 2013:23–51.
- Alvirez J, Greenwood GL, Johnson TL, Parker KL. *Intersectionality in Public Health Research: A View from the National Institutes of Health*. American Public Health Association; 2021:95–97.
- Braveman PA, Arkin E, Proctor D, Kauh T, Holm N. Systemic and structural racism: definitions, examples, health damages, and approaches to dismantling: study examines definitions, examples, health damages, and dismantling systemic and structural racism. *Health Aff*. 2022;41(2):171–178.
- Arsenault R. *Freedom Riders: 1961 and the Struggle for Racial Justice*. Oxford University Press; 2006.
- Williams DR, Collins C. Racial residential segregation: a fundamental cause of racial disparities in health. *Publ Health Rep*. 2001 Sep-Oct;116(5):404–416. <https://doi.org/10.1093/phr/116.5.404>. PMID: 12042604; PMCID: PMC1497358.
- Helper R. *Racial Policies and Practices of Real Estate Brokers*. U of Minnesota Press; 1969.
- Jackson K. *The cost of good intentions: the ghettoization of public housing in the United States*. New York: Crabgrass Frontier: Suburbanization United States. OxfordUniversity Press, Incorporated; 1985:219–396.
- Archer DN. White men’s roads through black men’s homes”: advancing racial equity through highway reconstruction. *Vand L Rev*. 2020;73:1259.
- Allen DJ. *Lost in the Transit Desert: Race, Transit Access, and Suburban Form*. Routledge; 2017.
- McLaughlin CG, Wyszewianski L. Access to care: remembering old lessons. *Health Serv Res*. Dec 2002;37(6):1441–1443. <https://doi.org/10.1111/1475-6773.12171>.
- Brown LT. *The Black Butterfly: The Harmful Politics of Race and Space in America*. JHU Press; 2021.
- Transit equity & environmental health in Baltimore city. <https://americanhealth.jhu.edu/sites/default/files/2021-09/JHU-016%20Transit%20Equity%20Report-FINAL.0.pdf>; 2021. Accessed April 5, 2024.
- Michael Dresser. "Hogan Says No to Red Line, Yes to Purple". Baltimore Sun.
- Colin Campbell. "Five Years Later, Many across Baltimore bitterly Lament Gov. Hogan’s Decision to Kill the Red Line Light Rail". Baltimore Sun.
- Lewis TA. *Working Definition of Ableism—January 2022 Update*. vol. 1. Talila A Lewis; 2022.
- Clarke P, Twardzik E, Meade MA, Peterson MD, Tate D. Social participation among adults aging with long-term physical disability: the role of socioenvironmental factors. *J Aging Health*. 2019;31(10_suppl):145S–168S.
- We Will Ride! – the Gang of 19; 2023. <https://adapt.org/we-will-ride-the-gang-of-19/>. Accessed July 18, 2023.
- American disabled for attendant programs Today ADAPT, In: American Disabled for Attendant Programs Today, editor. United States: Library of Congress.
- Bezyak JL, Sabella S, Hammel J, McDonald K, Jones RA, Barton D. Community participation and public transportation barriers experienced by people with disabilities. *Disabil Rehabil*. 2020;42(23):3275–3283.
- Brumbaugh S. *Travel patterns of American adults with disabilities*. Bureau of Transportation Statistics. Washington DC, WA, USA: US Department of Transportation; 2018.
- Winter M. *I Was There... Michael Winter Washington*; 1990. <https://adapt.org/g/1990-washington-michael-winter-ADAPT>, 1990.
- Eisenberg Y, Heider A, Gould R, Jones R. Are communities in the United States planning for pedestrians with disabilities? Findings from a systematic evaluation of local government barrier removal plans. *Cities*. 2020;102, 102720.
- Gupta D, Chen H-W, Miller LA, Surya F. Improving the efficiency of demand-responsive paratransit services. *Transport Res Pol Pract*. 2010;44(4):201–217.
- Bezyak JL, Sabella SA, Gattis RH. Public transportation: an investigation of barriers for people with disabilities. *J Disabil Pol Stud*. 2017;28(1):52–60.
- Egger SM, Gemperli A, Filippo M, Liechti R, Gantschnig BE. The experiences and needs of persons with disabilities in using paratransit services. *Disability Health J*. 2022;15(4), 101365.
- Desai RH, Choi S, Wehmeier A, Oxford JM, Morgan KA. Advocating for transportation equity: a critical examination of paratransit service reductions in St. Louis and its impact on health and community social participation. *Disability Health J*. 2024;17(4), 101666.
- Zimmerman R, Petrof C. Transportation delayed is transportation denied: report of on-time performance in Pace’s city of Chicago paratransit service. <https://www.accessliving.org/wp-content/uploads/2019/08/Final-Report-Transportation-Delayed-is-Transportation-Denied-10.22.17.pdf>; 2017. Accessed December 9, 2024.
- Hamraie A. Mapping access: digital humanities, disability justice, and sociospatial practice. *Am Q*. 2018;70(3):455–482.
- Hamraie A. Crip mobility justice: ableism and active transportation debates. *Int J Urban Reg Res*; 2021. <https://www.ijurr.org/spotlight-on/disabling-city/crip-mobility-justice/>. Accessed May 13, 2025.
- U.S. Census Bureau. Table B18101A: 2023 5-year Estimates Age by Disability Status (White Alone) & Table B18101B: 2023 5-year Estimates Age by Disability Status (Black or African American Alone). data.census.gov.
- Bailey M, Mobley IA. Work in the intersections: a black feminist disability framework. *Gend Soc*. 2019;33(1):19–40.
- Wright Marisa. A shared struggle for equality: disability rights and racial justice. <https://www.naacpldf.org/disability-rights-and-racial-justice/>; 2023. Accessed December 16, 2024.
- Perry David M, Carter-Long Lawrence. The Ruderman white paper on media coverage of law enforcement use of force and disability: a media study (2013-2015) and overview. <https://rudermanfoundation.org/media-missing-the-story-half-of-a-il-recent-high-profile-police-related-killings-are-people-with-disabilities/>; 2016. Accessed December 16, 2024.

40. Fuentes K, Hsu S, Patel S, Lindsay S. More than just double discrimination: a scoping review of the experiences and impact of ableism and racism in employment. *Disabil Rehabil.* 2024;46(4):650–671.
41. Wang Y, Shen Q, Ashour LA, Dannenberg AL. Ensuring equitable transportation for the disadvantaged: paratransit usage by persons with disabilities during the COVID-19 pandemic. *Transport Res Pol Pract.* 2022;159:84–95.
42. Nieuwenhuijsen MJ. Urban and transport planning pathways to carbon neutral, liveable and healthy cities; A review of the current evidence. *Environ Int.* 2020;140, 105661.
43. Heyer J, Palm M, Niemeier D. Are we keeping up? Accessibility, equity and air quality in regional planning. *J Transport Geogr.* 2020;89, 102891.
44. Schwartz N, Buliung R, Daniel A, Rothman L. Disability and pedestrian road traffic injury: a scoping review. *Health Place.* Sep 2022;77, 102896. <https://doi.org/10.1016/j.healthplace.2022.102896>.
45. Neuroth LM, Singichetti B, Harmon KJ, Waller AE, Naumann RB. Racial and ethnic disparities in motor vehicle crash-related outcomes in North Carolina surrounding the COVID-19 pandemic. *Inj Prev.* Jan 25 2024;30(1):84–88. <https://doi.org/10.1136/ip-2023-045005>.
46. Xiao C, Goryakin Y, Cecchini M. Physical activity levels and new public transit: a systematic review and meta-analysis. *Am J Prev Med.* Mar 2019;56(3):464–473. <https://doi.org/10.1016/j.amepre.2018.10.022>.
47. Palm M, Sturrock SL, Howell NA, Farber S, Widener MJ. The uneven impacts of avoiding public transit on riders' access to healthcare during COVID-19. *J Transport Health.* 2021;22, 101112.
48. Magnan S. *Social Determinants of Health 101 for Health Care: Five Plus Five.* NAM perspectives; 2017.
49. Litman T. *The Hidden Traffic Safety Solution: Public Transportation.* 2016.
50. Cidell J. *A Research Agenda for Transport Equity and Mobility Justice.* 2024.
51. Levine K, Karner A. Approaching accessibility: four opportunities to address the needs of disabled people in transportation planning in the United States. *Transp Policy.* 2023;131:66–74.
52. Jones Camara Phyllis, Ford Chandra L, Griffith Derek M, et al., eds. *Racism: Science & Tools for the Public Health Professional.* American Journal of Public Health; 2019.
53. Freire P. *Pedagogy of the Oppressed.* Bloomsbury publishing USA; 2018.
54. Lorde A. *Learning from the 60s. Sister Outsider: Essays and Speeches.* 1982, 13444.
55. Enright T. Transit justice as spatial justice: learning from activists. *Mobilities.* 2019; 14(5):665–680.
56. *Public involvement in transit decision-making. Federal Transit Administration;* 2023. <https://www.transit.dot.gov/Public-Involvement>. Accessed July 20, 2023.
57. O'Connor Rita, Schwartz Marcy, Schaad Joy, O'Connor Rita, Schwartz Marcy, Schaad Joy. State of the practice: white paper on public involvement. (A1D04): Committee on Public Involvement in Transportation. Transportation Research Board. <https://onlinepubs.trb.org/onlinepubs/millennium/00108.pdf>.
58. U.S. Department of Transportation. *Title VI Requirements and Guidelines for Federal Transit Administration Recipients.* Washington DC: Federal Transit Administration, FTA; 2012. editor. FTA C 47021B.